Unit 3

Chemical Kinetics

Chemical kinetics is the key to understanding how chemical reactions actually take place.

What this unit sets out to do

- define the average and the instantaneous rate of a reaction
- express a reaction's rate as the change in concentration of any reactant or product with time
- tell elementary reactions from complex ones
- distinguish the molecularity of a reaction from its order
- define the rate constant
- discuss the effect of concentration, temperature and catalysts on reaction rate
- derive the integrated rate equations of zero and first order reactions
- work out rate constants of zero and first order reactions
- explain the collision theory

Three questions about any reaction

- chemistry is about change: reactions turn substances with definite properties into others with different ones
- (a) **feasibility**, predicted by thermodynamics: at constant temperature and pressure a reaction with $\Delta G < 0$ is feasible
- (b) **extent**, found from chemical equilibrium
- (c) **speed**: the time a reaction takes to reach equilibrium

Why rate matters

- full understanding of a reaction needs its rate and the factors controlling it, as well as feasibility and extent
- what makes food spoil quickly? how can a fast-setting dental filling be designed? what governs how fast fuel burns in a car engine?

Chemical kinetics

- the branch of chemistry that studies reaction rates and their mechanisms, and answers such questions
- from the Greek *kinesis*, movement

Thermodynamics vs kinetics

- thermodynamics speaks only to feasibility; kinetics speaks to rate
- thermodynamic data say diamond should turn into graphite, yet the change is so slow as to be imperceptible, which is why most people think

What kinetics studies

- … that diamond is forever
- kinetics finds a reaction's speed and also the conditions under which that speed can be changed
- rate is affected by concentration, temperature, pressure and catalyst

Two levels of view

- **macroscopic**: amounts reacted or formed, and how fast they are used up or produced
- **molecular**: reaction mechanisms, i.e. the orientation and energy of colliding molecules

This unit covers the average and instantaneous rate of reaction and what affects them, plus some elementary ideas of the collision theory of reaction rates, starting with what reaction rate means.

3.1 Rate of a Chemical Reaction

Reactions differ widely in speed

- **very fast**: ionic reactions, e.g. silver chloride precipitates the instant aqueous silver nitrate and sodium chloride are mixed
- **very slow**: rusting of iron in air and moisture
- **moderate**: inversion of cane sugar; hydrolysis of starch
- try to name further examples of each kind

Defining rate

- a car's speed is change of position (distance covered) over a period of time
- likewise, the **rate of a reaction** is the change in concentration of a reactant or product per unit time
- stated as (i) the rate of fall in concentration of any one reactant, or (ii) the rate of rise in concentration of any one product

A model reaction, $R \rightarrow P$

- constant volume; one mole of R gives one mole of P
- $[R]_1$, $[P]_1$ at time t_1; $[R]_2$, $[P]_2$ at time t_2; square brackets mean molar concentration

$$\Delta t = t_2 - t_1 \quad \Delta[R] = [R]_2 - [R]_1 \quad \Delta[P] = [P]_2 - [P]_1$$

Rate of disappearance of R

$$\text{Rate of disappearance of R} = \frac{\text{decrease in concentration of R}}{\text{time taken}} = -\frac{\Delta[R]}{\Delta t}$$

Rate of appearance of P

$$\text{Rate of appearance of P} = \frac{\text{increase in concentration of P}}{\text{time taken}} = +\frac{\Delta[\text{P}]}{\Delta t}$$

- Δ[R] is negative, since reactant concentration is falling, so it is multiplied by –1 to keep the rate positive

Average rate, r_{av}

- the two expressions above give the **average rate of a reaction**, r_{av}
- it depends on how much the concentration of reactant or product changes and the time that change takes (Figure 3.A)

Figure 3.A Instantaneous and average rate of a reaction

- **(a)** concentration of reactant falls along a curve from $[R]_0$ as time passes
- the chord between $(t_1, [R]_1)$ and $(t_2, [R]_2)$ gives $r_{av} = -\Delta[R]/\Delta t = -([R]_2 - [R]_1)/(t_2 - t_1)$
- the tangent at time t gives $r_{inst} = -d[R]/dt = -\text{slope}$
- **(b)** concentration of product rises along a curve towards $[P]_\infty$
- the chord gives $r_{av} = \Delta[P]/\Delta t = ([P]_2 - [P]_1)/(t_2 - t_1)$
- the tangent gives $r_{inst} = d[P]/dt = \text{slope}$

Units of rate of a reaction

- from the definitions, rate has units of concentration time^{-1}
- concentration in mol L^{-1} and time in seconds: mol L^{-1} s^{-1}
- for gaseous reactions with concentrations given as partial pressures: atm s^{-1}

Worked example average rate of hydrolysis of butyl chloride

For $C_4H_9Cl + H_2O \rightarrow C_4H_9OH + HCl$, the concentration of C_4H_9Cl (butyl chloride) was followed over time. Find the average rate over the various time intervals.

t / s	0	50	100	150	200	300	400	700	800
[C_4H_9Cl] / mol L^{-1}	0.100	0.0905	0.0820	0.0741	0.0671	0.0549	0.0439	0.0210	0.017

- find the change in concentration over each interval and divide Δ[R] by Δt; the results are set out in Table 3.A

TABLE 3.A **Average rates of hydrolysis of butyl chloride**

$[C_4H_9Cl]_{t1}$ / mol L^{-1}	$[C_4H_9Cl]_{t2}$ / mol L^{-1}	t_1 / s	t_2 / s	$r_{av} \times 10^4$ / mol L^{-1} s^{-1} $= -\{([C_4H_9Cl]_{t2} - [C_4H_9Cl]_{t1})/(t_2 - t_1)\} \times 10^4$
0.100	0.0905	0	50	1.90
0.0905	0.0820	50	100	1.70
0.0820	0.0741	100	150	1.58
0.0741	0.0671	150	200	1.40
0.0671	0.0549	200	300	1.22
0.0549	0.0439	300	400	1.10
0.0439	0.0335	400	500	1.04
0.0210	0.017	700	800	0.4

Limits of the average rate

- in Table 3.A the average rate drops from 1.90 $\times 10^{-4}$ to 0.4 $\times 10^{-4}$ mol L^{-1} s^{-1}
- an average rate is constant over the interval it was calculated for, so it cannot give the rate at a particular instant

Instantaneous rate

- the rate at one moment: the average rate over the smallest possible interval, dt (Δt approaching zero)

$$r_{av} = \frac{-\Delta[R]}{\Delta t} = \frac{\Delta[P]}{\Delta t} \;;\; \text{as } \Delta t \to 0:$$

$$r_{inst} = \frac{-d[R]}{dt} = \frac{d[P]}{dt}$$

Figure 3.B Instantaneous rate of hydrolysis of butyl chloride(C_4H_9Cl)

- $[C_4H_9Cl]$ (M, from 0.1 down) against time (0 to 800 s): a smooth curve through the measured points, falling ever less steeply
- a tangent touches the curve at t = 600 s; the right-angled triangle drawn on it (from about 400 s to 800 s) gives its slope, the instantaneous rate at 600 s

Instantaneous rate from a graph

- draw a tangent at time t on the concentration–time curve for R or for P and find its slope (Figure 3.A)
- for the butyl chloride example, plot its concentration against time and draw the tangent that touches the curve at t = 600 s (Figure 3.B); its slope is the instantaneous rate

$$r_{inst} \text{ at } 600 \text{ s} = -\frac{0.0165 - 0.037}{(800 - 400)\ \text{s}}\ \text{mol L}^{-1} = \mathbf{5.12 \times 10^{-5}\ mol\ L^{-1}\ s^{-1}}$$

t / s	250	350	450
r_{inst} / mol L^{-1} s^{-1}	1.22×10^{-4}	1.0×10^{-4}	6.4×10^{-5}

Equal stoichiometric coefficients

$$Hg(l) + Cl_2(g) \rightarrow HgCl_2(s)$$

$$\text{Rate of reaction} = -\frac{\Delta[Hg]}{\Delta t} = -\frac{\Delta[Cl_2]}{\Delta t} = \frac{\Delta[HgCl_2]}{\Delta t}$$

- any reactant disappears at the same rate as the product appears

Unequal coefficients

$$2HI(g) \rightarrow H_2(g) + I_2(g)$$

- two moles of HI give one mole each of H_2 and I_2, so HI is used up twice as fast as H_2 or I_2 forms
- each rate is divided by its stoichiometric coefficient; here Δ[HI] is divided by 2:

$$\text{Rate of reaction} = -\frac{1}{2}\frac{\Delta[HI]}{\Delta t} = \frac{\Delta[H_2]}{\Delta t} = \frac{\Delta[I_2]}{\Delta t}$$

A further example

$$5Br^-(aq) + BrO_3^-(aq) + 6H^+(aq) \rightarrow 3Br_2(aq) + 3H_2O(l)$$

$$\text{Rate} = -\frac{1}{5}\frac{\Delta[Br^-]}{\Delta t} = -\frac{\Delta[BrO_3^-]}{\Delta t} = -\frac{1}{6}\frac{\Delta[H^+]}{\Delta t} = \frac{1}{3}\frac{\Delta[Br_2]}{\Delta t} = \frac{1}{3}\frac{\Delta[H_2O]}{\Delta t}$$

- for a gas reaction at constant temperature, concentration is directly proportional to partial pressure, so rate may also be given as the rate of change of a reactant's or product's partial pressure

Worked example average rate and rate of production

The decomposition of N_2O_5 in CCl_4 at 318 K is followed by measuring $[N_2O_5]$: it starts at 2.33 mol L^{-1} and falls to 2.08 mol L^{-1} after 184 minutes. The reaction is $2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$. Give the average rate per hour, per minute and per second, and the rate at which NO_2 is produced over this period.

$$\text{Average rate} = -\frac{1}{2}\frac{\Delta[N_2O_5]}{\Delta t} = -\frac{1}{2}\frac{(2.08 - 2.33)\ \text{mol L}^{-1}}{184\ \text{min}} = \mathbf{6.79 \times 10^{-4}\ mol\ L^{-1}\ min^{-1}}$$

- per hour: $(6.79 \times 10^{-4}\ \text{mol L}^{-1}\ \text{min}^{-1}) \times (60\ \text{min}/1\ \text{h}) = \mathbf{4.07 \times 10^{-2}\ mol\ L^{-1}/h}$
- per second: $6.79 \times 10^{-4}\ \text{mol L}^{-1} \times 1\ \text{min}/60\ \text{s} = \mathbf{1.13 \times 10^{-5}\ mol\ L^{-1}\ s^{-1}}$
- since rate = ¼ $\Delta[NO_2]/\Delta t$:

$$\frac{\Delta[NO_2]}{\Delta t} = 6.79 \times 10^{-4} \times 4 = \mathbf{2.72 \times 10^{-3}\ mol\ L^{-1}\ min^{-1}}$$

Intext questions

Q1 In R → P, the reactant concentration drops from 0.03M to 0.02M in 25 minutes. Find the average rate, with time in minutes and then in seconds.

Q2 In 2A → Products, [A] falls from 0.5 mol L^{-1} to 0.4 mol L^{-1} in 10 minutes. What is the rate over this interval?

3.2 Factors Influencing Rate of a Reaction

Reaction rate depends on the experimental conditions: reactant concentration (pressure, for gases), temperature and catalyst.

3.3 Dependence of Rate on Concentration

Rate law

- at a given temperature, rate may depend on the concentration of one or more reactants and products
- writing the rate in terms of reactant concentrations gives the **rate law**, also called the **rate equation** or **rate expression**

3.4 Rate Expression and Rate Constant

Rate depends on concentration

- Table 3.A shows the rate dropping as time passes and reactant concentrations fall
- conversely, rates usually rise when reactant concentrations rise; so rate depends on reactant concentration

General rate expression

- for $aA + bB \rightarrow cC + dD$ (a, b, c, d = stoichiometric coefficients):

$$\text{Rate} \propto [A]^x [B]^y \quad \text{or} \quad \boxed{\text{Rate} = k\,[A]^x [B]^y}$$

$$-\frac{d[R]}{dt} = k\,[A]^x [B]^y$$

- the exponents x and y may or may not equal a and b
- the last form is the **differential rate equation**; k is the proportionality constant, the **rate constant**

Rate law defined

- an equation relating reaction rate to reactant concentrations is a **rate law** or **rate expression**
- precisely: the rate written in terms of the molar concentrations of the reactants, each raised to a power that may or may not match that reactant's stoichiometric coefficient in the balanced equation

Finding a rate law: $2NO(g) + O_2(g) \rightarrow 2NO_2(g)$

- measure the rate for different initial concentrations, either varying one reactant while holding the other fixed, or varying both
- results are given in Table 3.B

Table 3.B Initial rate of formation of NO_2

Experiment	Initial [NO] / mol L^{-1}	Initial [O_2] / mol L^{-1}	Initial rate of formation of NO_2 / mol L^{-1} s^{-1}
1.	0.30	0.30	0.096
2.	0.60	0.30	0.384
3.	0.30	0.60	0.192
4.	0.60	0.60	0.768

Reading the data

- doubling [NO] with [O_2] fixed makes the initial rate **four** times larger (0.096 → 0.384 mol L^{-1} s^{-1}): rate depends on $[NO]^2$
- doubling [O_2] with [NO] fixed **doubles** the rate: rate depends on [O_2] to the first power
- so the rate equation is:

$$\boxed{\text{Rate} = k\,[NO]^2[O_2]}$$

Differential form for $NO + O_2$

$$-\frac{d[R]}{dt} = k\,[NO]^2[O_2]$$

- in this experimentally derived rate equation, the exponents happen to equal the stoichiometric coefficients of the balanced equation

Rate laws must be measured

- in the reactions below the exponents do **not** match the stoichiometric coefficients
- so a rate law cannot be predicted in theory just by looking at the balanced equation; it has to be found by experiment

Reaction	Rate expression found by experiment
1. $CHCl_3 + Cl_2 \rightarrow CCl_4 + HCl$	Rate = $k\,[CHCl_3]\,[Cl_2]^{1/2}$
2. $CH_3COOC_2H_5 + H_2O \rightarrow CH_3COOH + C_2H_5OH$	Rate = $k\,[CH_3COOC_2H_5]^1\,[H_2O]^0$

3.5 Order of a Reaction

Order defined

- in Rate = $k\,[A]^x\,[B]^y$, x and y show how sensitive the rate is to changes in [A] and [B]
- x and y are the orders with respect to A and B; their sum $x + y$ is the **overall order**
- the **order** of a reaction is the sum of the powers of the reactant concentrations in the rate law

Possible values

- order may be 0, 1, 2, 3, or even a fraction
- **zero order**: the rate does not depend on reactant concentrations at all

Worked example overall order

Find the overall order for (a) Rate = $k\,[A]^{1/2}\,[B]^{3/2}$ and (b) Rate = $k\,[A]^{3/2}\,[B]^{-1}$.

- for Rate = $k\,[A]^x\,[B]^y$, order = $x + y$
- (a) order = 1/2 + 3/2 = **2**: second order
- (b) order = 3/2 + (–1) = **1/2**: half order

Elementary and complex reactions

- a balanced equation never shows how a reaction really happens, since reactions seldom finish in a single step
- **elementary reactions**: those that take place in one step
- **complex reactions**: those in which a sequence of elementary reactions (the **mechanism**) yields the products

Kinds of complex reaction

- **consecutive** reactions, e.g. oxidation of ethane to CO_2 and H_2O runs through intermediate steps forming an alcohol, an aldehyde and an acid
- **reverse** reactions
- **side** reactions, e.g. nitration of phenol gives both o-nitrophenol and p-nitrophenol

Units of rate constant

- for $aA + bB \rightarrow cC + dD$ with Rate = $k\,[A]^x\,[B]^y$ and order $n = x + y$:

$$k = \frac{\text{Rate}}{[A]^x[B]^y} = \frac{\text{concentration}}{\text{time}} \times \frac{1}{(\text{concentration})^n}$$

- with concentration in mol L^{-1} and time in s (SI), the units for each order are in Table 3.C

Table 3.C Units of rate constant

Reaction	Order	Units of k
Zero order	0	$\frac{\text{mol L}^{-1}}{\text{s}} \times \frac{1}{(\text{mol L}^{-1})^0} = \text{mol L}^{-1}\,\text{s}^{-1}$
First order	1	$\frac{\text{mol L}^{-1}}{\text{s}} \times \frac{1}{(\text{mol L}^{-1})^1} = \text{s}^{-1}$
Second order	2	$\frac{\text{mol L}^{-1}}{\text{s}} \times \frac{1}{(\text{mol L}^{-1})^2} = \text{mol}^{-1}\,\text{L}\,\text{s}^{-1}$

Worked example order from the units of k

State the order of reaction for each rate constant: (i) $k = 2.3 \times 10^{-5}$ L mol^{-1} s^{-1}; (ii) $k = 3 \times 10^{-4}$ s^{-1}.

- (i) L mol^{-1} s^{-1} is the unit of a second order rate constant, so this is a **second order** reaction
- (ii) s^{-1} is the unit of a first order rate constant, so this is a **first order** reaction

3.6 Molecularity of a Reaction

Definition

- a further property that helps in understanding mechanism
- **molecularity**: the number of reacting species (atoms, ions or molecules) in an elementary reaction that must collide at the same moment for reaction to occur

Unimolecular

- one reacting species involved
- e.g. decomposition of ammonium nitrite

Unimolecular

$NH_4NO_2 \rightarrow N_2 + 2H_2O$

- one species reacts

Bimolecular

$2HI \rightarrow H_2 + I_2$

- two species collide together, e.g. dissociation of hydrogen iodide

Trimolecular

$2NO + O_2 \rightarrow 2NO_2$

- (termolecular) three species collide together

Beyond three

- the chance of more than three molecules colliding and reacting at once is very small, so molecularity-three reactions are very rare and slow
- so any complex reaction whose stoichiometric equation involves over three molecules has to proceed in several steps
- $KClO_3 + 6FeSO_4 + 3H_2SO_4 \rightarrow KCl + 3Fe_2(SO_4)_3 + 3H_2O$ looks tenth order but is actually **second order**, so it proceeds in several steps

Rate determining step

- which step controls the overall rate? The mechanism tells us
- a relay team's chance of winning hangs on its slowest runner; likewise the overall rate is controlled by the slowest step, the **rate determining step**

Example: H_2O_2 decomposition catalysed by iodide in alkaline medium

$$2H_2O_2 \rightarrow 2H_2O + O_2 \quad (I^-, \text{alkaline medium})$$

$$\text{Rate} = -\frac{d[H_2O_2]}{dt} = k\,[H_2O_2][I^-]$$

- first order in both H_2O_2 and I^-; evidence points to two steps:

(1) $H_2O_2 + I^- \rightarrow H_2O + IO^-$
(2) $H_2O_2 + IO^- \rightarrow H_2O + I^- + O_2$

- both steps are bimolecular elementary reactions
- IO^- is an **intermediate**: formed during the reaction but absent from the overall balanced equation
- step (1) is slow and so is the rate determining step; the rate at which the intermediate forms sets the rate of the reaction

Order vs molecularity: conclusions

- (i) order is found by experiment and may be zero or even a fraction; molecularity can never be zero or a non-integer
- (ii) order applies to elementary and complex reactions; molecularity only to elementary ones, and has no meaning for a complex reaction

(iii) For a complex reaction, the slowest step fixes the order: the overall order equals the molecularity of that slowest step.

Intext questions

Q3 For A + B → Product the rate law is $r = k\,[A]^{1/2}\,[B]^{2}$. What is the order of the reaction?

Q4 X turns into Y by second order kinetics. How does tripling the concentration of X affect the rate at which Y forms?

3.7 Integrated Rate Equations

Why integrate?

- the concentration dependence of rate is the differential rate equation
- instantaneous rate is awkward to find, needing the slope of a tangent at time *t* on the concentration–time plot (Figure 3.A); this makes the rate law, and so the order, hard to determine
- integrating the differential equation instead links the rate constant to directly measured data: concentrations at different times

Scope

- each reaction order has its own integrated rate equation
- here they are derived only for zero and first order reactions

3.8 Zero Order Reactions

- rate proportional to the **zero** power of reactant concentration; for R → P:

$$\text{Rate} = -\frac{d[R]}{dt} = k[R]^{0} \;;\; \text{any quantity to the power zero is 1, so } \text{Rate} = -\frac{d[R]}{dt} = k \times 1$$

$$d[R] = -k\,dt$$

- integrating both sides (*I* = constant of integration):

$$[R] = -kt + I$$

- at $t = 0$, $[R] = [R]_0$, the initial concentration: $[R]_0 = -k \times 0 + I$, so $[R]_0 = I$

$$\boxed{[R] = -kt + [R]_0}$$

The zero order plot

- $[R] = -kt + [R]_0$ has the straight-line form $y = mx + c$
- plotting [R] against t gives a straight line (Figure 3.C): slope = $-k$, intercept = $[R]_0$
- rearranged for the rate constant:

$$k = \frac{[R]_0 - [R]}{t}$$

Figure 3.C Variation in the concentration vs time plot for a zero order reaction

- concentration of R against time: a straight line falling from $[R]_0$ on the vertical axis
- its slope equals $-k$

Where zero order occurs

- fairly uncommon; seen under special conditions
- certain reactions catalysed by enzymes, and reactions taking place on metal surfaces
- thermal decomposition of HI on a gold surface

NH_3 on hot platinum

$2NH_3(g) \rightarrow N_2(g) + 3H_2(g)$ (1130 K, Pt catalyst)

$$\text{Rate} = k\,[NH_3]^0 = k$$

- zero order at high pressure; platinum is the catalyst
- at high pressure the metal surface is saturated with gas molecules, so changing conditions further cannot change the amount of NH_3 on the surface; the rate no longer depends on its concentration

3.9 First Order Reactions

- rate is proportional to the **first** power of the concentration of R; for R → P:

$$\text{Rate} = -\frac{d[R]}{dt} = k[R] \quad \text{or} \quad \frac{d[R]}{[R]} = -k\,dt$$

- integrating (I = constant of integration, easily found):

$$\ln [R] = -kt + I$$

- at $t = 0$, $[R] = [R]_0$: $\ln [R]_0 = -k \times 0 + I$, so $\ln [R]_0 = I$

$$\ln [R] = -kt + \ln [R]_0$$

Rearranged forms

$$\ln \frac{[R]}{[R]_0} = -kt \quad \text{or} \quad \boxed{k = \frac{1}{t} \ln \frac{[R]_0}{[R]}}$$

- taking antilogs of $\ln([R]/[R]_0) = -kt$:

$$\boxed{[R] = [R]_0 \, e^{-kt}}$$

From two times

- with $[R]_1$ at t_1 and $[R]_2$ at t_2:

$$\ln[R]_1 = -kt_1 + \ln[R]_0 \;;\; \ln[R]_2 = -kt_2 + \ln[R]_0$$

- subtracting the second from the first: $\ln[R]_1 - \ln[R]_2 = -kt_1 - (-kt_2)$

$$\ln \frac{[R]_1}{[R]_2} = k(t_2 - t_1) \Rightarrow \boxed{k = \frac{1}{(t_2 - t_1)} \ln \frac{[R]_1}{[R]_2}}$$

Straight-line plots

- $\ln [R] = -kt + \ln [R]_0$ matches $y = mx + c$: $\ln [R]$ against t (Figure 3.D) is a straight line with slope $-k$ and intercept $\ln [R]_0$
- in base-10 logarithms:

$$\boxed{k = \frac{2.303}{t} \log \frac{[R]_0}{[R]}} \quad \text{so} \quad \log \frac{[R]_0}{[R]} = \frac{kt}{2.303}$$

- $\log([R]_0/[R])$ against t (Figure 3.E) has slope $k/2.303$

Examples of first order

- hydrogenation of ethene: $C_2H_4(g) + H_2(g) \rightarrow C_2H_6(g)$, with Rate = $k\,[C_2H_4]$
- radioactive decay of unstable nuclei, natural or artificial, always follows first order kinetics

For ln and log (logarithms), see Appendix IV of the book.

Figure 3.D A plot between ln[R] and *t* for a first order reaction

- ln[R] against time: a straight line falling from $\ln[R]_0$ on the vertical axis, with slope $-k$

Figure 3.E Plot of log $[R]_0/[R]$ vs time for a first order reaction

- $\log([R]_0/[R])$ against time: a straight line rising from the origin, with slope $k/2.303$

More first order examples

$$^{226}_{88}\text{Ra} \rightarrow {}^{4}_{2}\text{He} + {}^{222}_{86}\text{Rn} \qquad \text{Rate} = k\,[\text{Ra}]$$

- radioactive decay of radium, as above
- decomposition of N_2O_5 and of N_2O

Worked example first order rate constant

For the first order reaction $N_2O_5(g) \rightarrow 2NO_2(g) + ½O_2(g)$ at 318 K, $[N_2O_5]$ is 1.24×10^{-2} mol L^{-1} at the start and 0.20×10^{-2} mol L^{-1} after 60 minutes. Find the rate constant at 318 K.

$$\log \frac{[R]_1}{[R]_2} = \frac{k(t_2 - t_1)}{2.303} \Rightarrow k = \frac{2.303}{(t_2 - t_1)} \log \frac{[R]_1}{[R]_2}$$

$$= \frac{2.303}{(60\text{ min} - 0\text{ min})} \log \frac{1.24 \times 10^{-2}\text{ mol L}^{-1}}{0.20 \times 10^{-2}\text{ mol L}^{-1}} = \frac{2.303}{60} \log 6.2\ \text{min}^{-1} \Rightarrow \mathbf{k = 0.0304\ min^{-1}}$$

First order gas-phase reaction

- take a typical first order gas reaction $A(g) \rightarrow B(g) + C(g)$
- p_i = initial pressure of A; p_t = total pressure at time '*t*'; the integrated rate equation is derived from:

$$\text{total pressure } p_t = p_A + p_B + p_C \text{ (pressure units)}$$

Pressure form of the first order equation

- p_A, p_B, p_C = partial pressures of A, B and C
- if A's pressure falls by x atm by time t, and one mole each of B and C forms, then B and C each rise by x atm

	A(g)	→ B(g)	+ C(g)
at $t = 0$	p_i atm	0 atm	0 atm
at time t	$(p_i - x)$ atm	x atm	x atm

- p_i = initial pressure at $t = 0$

$$p_t = (p_i - x) + x + x = p_i + x \;\Rightarrow\; x = (p_t - p_i)$$

$$p_A = p_i - x = p_i - (p_t - p_i) = 2p_i - p_t$$

$$k = \frac{2.303}{t} \log \frac{p_i}{p_A} = \frac{2.303}{t} \log \frac{p_i}{(2p_i - p_t)}$$

Worked example rate constant from total pressure (part 1)

$N_2O_5(g)$ decomposes thermally by first order kinetics at constant volume: $2N_2O_5(g) \rightarrow 2N_2O_4(g) + O_2(g)$. The total pressure is 0.5 atm at 0 s and 0.512 atm at 100 s. Find the rate constant.

- let N_2O_5 pressure fall by $2x$ atm; two moles of N_2O_5 give two of N_2O_4 and one of O_2, so N_2O_4 rises by $2x$ atm and O_2 by x atm

	$2N_2O_5(g)$	→ $2N_2O_4(g)$	+ $O_2(g)$
start, $t = 0$	0.5 atm	0 atm	0 atm
at time t	$(0.5 - 2x)$ atm	$2x$ atm	x atm

$$p_t = p_{N_2O_5} + p_{N_2O_4} + p_{O_2} = (0.5 - 2x) + 2x + x = 0.5 + x \;\Rightarrow\; x = p_t - 0.5$$

$$p_{N_2O_5} = 0.5 - 2x = 0.5 - 2(p_t - 0.5) = 1.5 - 2p_t \;;\; \text{at } t = 100 \text{ s}, p_t = 0.512 \text{ atm}$$

Worked example rate constant from total pressure (part 2)

$$p_{N_2O_5} = 1.5 - 2 \times 0.512 = 0.476 \text{ atm}$$

$$k = \frac{2.303}{t} \log \frac{p_i}{p_A} = \frac{2.303}{100 \text{ s}} \log \frac{0.5 \text{ atm}}{0.476 \text{ atm}} = \frac{2.303}{100 \text{ s}} \times 0.0216 = \mathbf{4.98 \times 10^{-4} \text{ s}^{-1}}$$

3.10 Half-Life of a Reaction

The **half-life** of a reaction, $t_{1/2}$, is the time taken for a reactant's concentration to fall to half its initial value.

Zero order half-life

- start from $k = ([R]_0 - [R])/t$; at $t = t_{1/2}$, $[R] = ½[R]_0$:

$$k = \frac{[R]_0 - ½[R]_0}{t_{1/2}} \Rightarrow \boxed{t_{1/2} = \frac{[R]_0}{2k}}$$

- directly proportional to the initial reactant concentration and inversely proportional to the rate constant

First order half-life

- start from $k = (2.303/t) \log([R]_0/[R])$; at $t_{1/2}$, $[R] = [R]_0/2$:

$$k = \frac{2.303}{t_{1/2}} \log \frac{[R]_0}{[R]_0/2} \Rightarrow t_{1/2} = \frac{2.303}{k} \log 2$$

$$t_{1/2} = \frac{2.303}{k} \times 0.301 \Rightarrow \boxed{t_{1/2} = \frac{0.693}{k}}$$

Worked example half-life from k

A first order reaction has $k = 5.5 \times 10^{-14}\ s^{-1}$. What is its half-life?

$$t_{1/2} = \frac{0.693}{k} = \frac{0.693}{5.5 \times 10^{-14}\ s^{-1}} = \mathbf{1.26 \times 10^{13}\ s}$$

Worked example 99.9% completion

Prove that a first order reaction takes 10 times its half-life ($t_{1/2}$) to reach 99.9% completion.

- at 99.9% completion, $[R] = [R]_0 - 0.999[R]_0$

$$k = \frac{2.303}{t} \log \frac{[R]_0}{[R]_0 - 0.999[R]_0} = \frac{2.303}{t} \log 10^3$$

$$t = 6.909/k \ ; \ t_{1/2} = 0.693/k$$

$$\frac{t}{t_{1/2}} = \frac{6.909}{k} \times \frac{k}{0.693} = \mathbf{10}$$

First order half-life is constant

- for a first order reaction $t_{1/2}$ does not depend on the initial concentration of the reactant
- half-life follows easily from the rate constant, and the rate constant from the half-life

Comparison

- zero order: $t_{1/2} \propto [R]_0$
- first order: $t_{1/2}$ independent of $[R]_0$
- the mathematical features of the zero and first order integrated laws are collected in Table 3.D

TABLE 3.D *Integrated Rate Laws for the Reactions of Zero and First Order*

Order	Reaction type	Differential rate law	Integrated rate law	Straight-line plot	Half-life	Units of k
0	$R \rightarrow P$	$d[R]/dt = -k$	$kt = [R]_0 - [R]$	$[R]$ vs t	$[R]_0/2k$	conc $time^{-1}$ or mol L^{-1} s^{-1}
1	$R \rightarrow P$	$d[R]/dt = -k[R]$	$[R] = [R]_0 e^{-kt}$ or $kt = \ln\{[R]_0/[R]\}$	$\ln[R]$ vs t	$\ln 2/k$	$time^{-1}$ or s^{-1}

Pseudo first order reactions

- conditions can sometimes change a reaction's order: many reactions of higher order obey a first order rate law
- hydrolysis of ethyl acetate (ethyl acetate + water) is truly **second order**, with both concentrations affecting the rate
- but water is used in large excess, so its concentration barely changes and only [ethyl acetate] affects the rate; e.g. 0.01 mol of ethyl acetate with 10 mol of water:

	$CH_3COOC_2H_5$	+ H_2O	→ CH_3COOH (H^+)	+ C_2H_5OH
$t = 0$	0.01 mol	10 mol	0 mol	0 mol
t (completion)	0 mol	9.99 mol	0.01 mol	0.01 mol

- since [H_2O] hardly changes, the reaction behaves as first order: a **pseudo first order reaction**
- inversion of cane sugar is another; Rate = k [$C_{12}H_{22}O_{11}$]:

$$C_{12}H_{22}O_{11}\text{ (cane sugar)} + H_2O \rightarrow C_6H_{12}O_6\text{ (glucose)} + C_6H_{12}O_6\text{ (fructose)} \quad (H^+)$$

Intext questions

Q5 A first order reaction has $k = 1.15 \times 10^{-3}$ s^{-1}. How long does it take 5 g of the reactant to fall to 3 g?

Q6 SO_2Cl_2 takes 60 minutes to decompose to half its starting amount. Treating the decomposition as first order, find its rate constant.

3.11 Temperature Dependence of the Rate of a Reaction

Heat speeds most reactions

- decomposition of N_2O_5: half the starting material decomposes in 12 min at 50 °C, 5 h at 25 °C and 10 days at 0 °C
- $KMnO_4$ mixed with oxalic acid ($H_2C_2O_4$) loses its colour faster when warmer
- a rise of 10° roughly doubles the rate constant

Arrhenius equation

- accounts accurately for how rate depends on temperature
- first put forward by the Dutch chemist J.H. van't Hoff; the Swedish chemist Arrhenius supplied its physical justification and interpretation

The Arrhenius equation

$$k = A\, e^{-E_a/RT}$$

- A = **Arrhenius factor** or **frequency factor** (also pre-exponential factor), a constant specific to each reaction
- R = gas constant; E_a = activation energy, in joules/mole (J mol^{-1})

$H_2(g) + I_2(g) \rightarrow 2HI(g)$

- by Arrhenius's picture, reaction happens only when an H_2 molecule and an I_2 molecule collide to form an unstable intermediate (Figure 3.F)
- the intermediate lasts a very short time, then breaks up into two HI molecules

Figure 3.F Formation of HI through the intermediate

- H–H and I–I approach side by side; in the intermediate, each H is partly bonded to an I (H···I) while the H···H and I···I links weaken
- the intermediate separates into two H–I molecules

Activation energy

- the energy needed to form this intermediate, the **activated complex** (C), is the **activation energy** (E_a)
- plotting potential energy against **reaction coordinate** (the energy profile as reactants become products) gives Figure 3.G
- some energy is released as the complex breaks into products, so the final enthalpy of reaction depends on the nature of reactants and products

Figure 3.G Diagram showing plot of potential energy vs reaction coordinate

- reactants $H_2 + I_2$ (A) climb over an energy hump whose peak is the activated complex C; the climb is the activation energy
- products 2HI (B) sit lower than the reactants; the difference is ΔH

Spread of molecular energies

- molecules do not all share the same kinetic energy; one molecule's behaviour cannot be predicted precisely, so Ludwig Boltzmann and James Clark Maxwell treated large populations of molecules statistically
- plot the fraction of molecules (N_E/N_T) with kinetic energy E against E (Figure 3.H); N_E = molecules with energy E, N_T = total molecules
- the peak is the **most probable kinetic energy**, the energy of the largest fraction; fewer molecules have energies above or below it

Figure 3.H Distribution curve showing energies among gaseous molecules

- fraction of molecules against kinetic energy: the curve rises from zero to a single peak (the most probable kinetic energy) and then tails off slowly towards high energies

Raising the temperature

- the peak of the curve moves to higher energy (Figure 3.I) and the curve broadens, spreading to the right, so a larger share of molecules has much higher energy
- the area under the curve stays constant, since total probability is always one
- E_a can be marked on the Maxwell–Boltzmann curve

Figure 3.I Distribution curve showing temperature dependence of rate of a reaction

- two curves, at t and at $(t + 10)$: the higher-temperature curve is lower-peaked, shifted right and more spread out
- beyond the marked energy of activation, a shaded tail shows the fraction reacting at t, and an extra area shows the additional molecules able to react at $(t + 10)$

Why rate doubles

- heating increases the fraction of molecules that collide with energies above E_a
- in the $(t + 10)$ curve, the area for molecules with energy $\geq E_a$ doubles, and so the rate doubles
- in the Arrhenius equation, the factor $e^{-E_a/RT}$ is the fraction of molecules with kinetic energy above E_a

Logarithmic form

- natural logs of both sides of $k = A\, e^{-E_a/RT}$:

$$\ln k = -\frac{E_a}{RT} + \ln A$$

- $\ln k$ against $1/T$ is a straight line (Figure 3.J): slope $= -E_a/R$, intercept $= \ln A$, from which E_a and A can be found

Figure 3.J A plot between $\ln k$ and $1/T$

- $\ln k$ against $1/T$: a straight line sloping downward with slope $-E_a/R$ and intercept $\ln A$

At two temperatures

- k_1, k_2 = rate constants at T_1, T_2; A is constant for a given reaction:

$$\ln k_1 = -\frac{E_a}{RT_1} + \ln A \;;\; \ln k_2 = -\frac{E_a}{RT_2} + \ln A$$

Conclusion from the Arrhenius equation

- raising the temperature, or lowering the activation energy, increases the rate of reaction
- and makes the rate constant rise exponentially

Arrhenius equation for two temperatures

- subtracting the equation at T_1 from that at T_2:

$$\ln k_2 - \ln k_1 = \frac{E_a}{RT_1} - \frac{E_a}{RT_2} \Rightarrow \ln \frac{k_2}{k_1} = \frac{E_a}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right) = \frac{E_a}{2.303R}\left[\frac{T_2 - T_1}{T_1 T_2}\right]$$

Worked example E_a and A from two rate constants

A reaction has rate constants 0.02 s^{-1} at 500 K and 0.07 s^{-1} at 700 K. Find E_a and A.

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303R}\left[\frac{T_2 - T_1}{T_1 T_2}\right] \Rightarrow \log \frac{0.07}{0.02} = \frac{E_a}{2.303 \times 8.314\ \text{J K}^{-1}\ \text{mol}^{-1}}\left[\frac{700 - 500}{700 \times 500}\right]$$

$$0.544 = E_a \times 5.714 \times 10^{-4}/19.15 \Rightarrow E_a = 0.544 \times 19.15/5.714 \times 10^{-4} = \mathbf{18230.8\ J}$$

$$\text{from } k = Ae^{-E_a/RT}\text{: } 0.02 = Ae^{-18230.8/8.314 \times 500} \Rightarrow A = 0.02/0.012 = \mathbf{1.61}$$

Worked example rate constant at a new temperature (part 1)

For $C_2H_5I(g) \rightarrow C_2H_4(g) + HI(g)$, the first order rate constant at 600 K is 1.60×10^{-5} s^{-1} and the activation energy is 209 kJ/mol. Find the rate constant at 700 K.

$$\log k_2 - \log k_1 = \frac{E_a}{2.303R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

Worked example rate constant at a new temperature (part 2)

$$\log k_2 = \log k_1 + \frac{E_a}{2.303R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right) = \log(1.60 \times 10^{-5}) + \frac{209000\ \text{J mol}^{-1}}{2.303 \times 8.314\ \text{J mol}^{-1}\ \text{K}^{-1}}\left(\frac{1}{600\ \text{K}} - \frac{1}{700\ \text{K}}\right)$$

$$\log k_2 = -4.796 + 2.599 = -2.197 \Rightarrow \mathbf{k_2 = 6.36 \times 10^{-3}\ s^{-1}}$$

3.12 Effect of Catalyst

Catalysts and inhibitors

- **catalyst**: a substance that speeds up a reaction while remaining, in the end, chemically unchanged itself
- e.g. MnO_2 greatly speeds up $2KClO_3 \rightarrow 2KCl + 3O_2$
- a substance that **slows** a reaction should not be called a catalyst; it is an **inhibitor**

Intermediate complex theory

- the catalyst joins in by bonding temporarily to the reactants, which gives an intermediate complex
- the complex exists only briefly, then breaks down into products and the catalyst

Lower activation energy

- the catalyst is believed to offer an alternative pathway (mechanism) with lower activation energy between reactants and products, lowering the potential energy barrier (Figure 3.K)
- by the Arrhenius equation, the lower the activation energy, the faster the reaction

Figure 3.K Effect of catalyst on activation energy

- potential energy against reaction coordinate: from the same reactant level to the same product level, the uncatalysed path climbs a high barrier while the catalysed path crosses a lower one

What a catalyst does not do

- a small amount of catalyst can act on a large amount of reactant
- it leaves the Gibbs energy, ΔG, unchanged: it catalyses spontaneous reactions but not non-spontaneous ones
- it does not change the equilibrium constant; it speeds forward and backward reactions equally, so the same equilibrium is simply reached sooner

3.13 Collision Theory of Chemical Reactions

- the Arrhenius equation holds widely, but **collision theory**, developed by Max Trautz and William Lewis in 1916–18, gives deeper insight into the energetics and mechanism of reactions
- it rests on the kinetic theory of gases

Basic assumptions

- reactant molecules are treated as hard spheres; reaction occurs when they collide
- **collision frequency** (Z): the number of collisions each second in each unit volume of the reacting mixture
- activation energy is the other factor affecting rate

Rate from collisions

- for a bimolecular elementary reaction A + B → Products:

$$\text{Rate} = Z_{AB}\, e^{-E_a/RT}$$

- Z_{AB} = collision frequency of A and B; $e^{-E_a/RT}$ = fraction of molecules with energy ≥ E_a
- comparison with the Arrhenius equation shows A is related to collision frequency

Effective collisions

- the equation predicts rate constants fairly well for atoms or simple molecules, but complex molecules deviate markedly, perhaps because not every collision gives products
- **effective collisions**: those with enough kinetic energy (the **threshold energy***) and the right orientation to break bonds between reacting species and form the new bonds of the products
- e.g. forming methanol from bromomethane depends on how the molecules are oriented (Figure 3.L): proper orientation leads to bond formation; improper orientation makes them just bounce apart with no product

Figure 3.L Diagram showing molecules having proper and improper orientation

$$CH_3Br + OH^- \rightarrow CH_3OH + Br^-$$

- **improper**: OH^- approaches the Br end of CH_3Br; the species rebound and give no products
- **proper**: OH^- approaches the carbon from the side opposite Br, forming an intermediate HO···C···Br, which gives CH_3OH and Br^-

Steric factor, P

- to allow for effective collisions, a **probability** or **steric factor** P is added, recognising that colliding molecules must be properly oriented:

$$\text{Rate} = P\, Z_{AB}\, e^{-E_a/RT}$$

- so activation energy together with proper orientation decide whether a collision is effective, and hence the rate

Limitations

- it treats atoms and molecules as hard spheres and ignores their structure
- fuller treatments and other theories come in higher classes

* Threshold energy is the activation energy plus the energy the reacting species already have.

Intext questions

Q7 How does temperature affect the rate constant?

Q8 Raising the absolute temperature by 10 K from 298 K doubles the rate of a reaction. Find E_a.

Q9 For $2HI(g) \rightarrow H_2 + I_2(g)$, the activation energy is 209.5 kJ mol^{-1} at 581 K. What fraction of reactant molecules has energy equal to or greater than the activation energy?

Summary

Kinetics and rate

- chemical kinetics studies reactions in terms of their rates, the effect of various variables, the rearrangement of atoms and the formation of intermediates
- rate = fall in reactant concentration, or rise in product concentration, per unit time; stated as an instantaneous rate at one moment or an average rate over a long interval
- temperature, reactant concentration and catalyst all affect rate

Rate law, order, rate constant

- the rate law expresses rate mathematically; it must be found by experiment, not predicted
- order with respect to a reactant = the power of its concentration in the rate law; overall order = the sum of these powers
- the rate constant is the proportionality factor in the rate law; it and the order can be found from the rate law or its integrated form

Molecularity

- defined only for elementary reactions, with values from 1 to 3; order may be 0, 1, 2, 3 or a fraction
- for an elementary reaction, molecularity and order are the same

Temperature, catalysts, collisions

- Arrhenius equation $k = Ae^{-E_a/RT}$: E_a = energy gap between activated complex and reactants; A (Arrhenius or pre-exponential factor) relates to collision frequency
- higher temperature or lower E_a raises the rate; a catalyst lowers E_a by offering another path
- collision theory adds the steric factor P (orientation of colliding molecules), which contributes to effective collisions: $k = P\,Z_{AB}\,e^{-E_a/RT}$

Exercises

E1 Using each rate expression, find the order of reaction and the dimensions of the rate constant: (i) $3NO(g) \rightarrow N_2O(g)$, Rate = $k[NO]^2$; (ii) $H_2O_2(aq) + 3I^-(aq) + 2H^+ \rightarrow 2H_2O(l) + I_3^-$, Rate = $k[H_2O_2][I^-]$; (iii) $CH_3CHO(g) \rightarrow CH_4(g) + CO(g)$, Rate = $k[CH_3CHO]^{3/2}$; (iv) $C_2H_5Cl(g) \rightarrow C_2H_4(g) + HCl(g)$, Rate = $k[C_2H_5Cl]$.

E2 For $2A + B \rightarrow A_2B$, rate = $k[A][B]^2$ with $k = 2.0 \times 10^{-6}$ mol^{-2} L^2 s^{-1}. Find the initial rate when [A] = 0.1 mol L^{-1} and [B] = 0.2 mol L^{-1}, and the rate once [A] has fallen to 0.06 mol L^{-1}.

E3 NH_3 decomposes on a platinum surface by a zero order reaction. With $k = 2.5 \times 10^{-4}$ mol^{-1} L s^{-1}, at what rates are N_2 and H_2 produced?

E4 Dimethyl ether decomposes to CH_4, H_2 and CO with Rate = $k[CH_3OCH_3]^{3/2}$. The reaction is followed by the pressure rise in a closed vessel, so the rate may also be written in terms of the partial pressure of dimethyl ether: Rate = $k(p_{CH_3OCH_3})^{3/2}$. With pressure in bar and time in minutes, what are the units of the rate and of the rate constant?

E5 List the factors that influence the rate of a chemical reaction.

E6 A reaction is second order in one reactant. How does its rate change if that reactant's concentration is (i) doubled; (ii) halved?

E7 How does temperature affect the rate constant of a reaction, and how can this effect be expressed quantitatively?

E8 A pseudo first order reaction in water gave these results. Find the average rate between 30 and 60 seconds.

t / s	0	30	60	90
[A] / mol L^{-1}	0.55	0.31	0.17	0.085

E9 A reaction is first order in A and second order in B. (i) Write its differential rate equation. (ii) What happens to the rate when [B] is tripled? (iii) What happens when both [A] and [B] are doubled?

Exercises continued

E10 For a reaction between A and B, the initial rate r_0 was measured at several starting concentrations. What is the order with respect to A and to B?

A / mol L^{-1}	0.20	0.20	0.40
B / mol L^{-1}	0.30	0.10	0.05
r_0 / mol L^{-1} s^{-1}	5.07×10^{-5}	5.07×10^{-5}	1.43×10^{-4}

E11 Kinetic study of 2A + B → C + D gave the results below. Find the rate law and the rate constant.

Experiment	**[A] / mol L^{-1}**	**[B] / mol L^{-1}**	**Initial rate of formation of D / mol L^{-1} min^{-1}**
I	0.1	0.1	6.0×10^{-3}
II	0.3	0.2	7.2×10^{-2}
III	0.3	0.4	2.88×10^{-1}
IV	0.4	0.1	2.40×10^{-2}

E12 A reaction between A and B is first order in A and zero order in B. Complete the table.

Experiment	**[A] / mol L^{-1}**	**[B] / mol L^{-1}**	**Initial rate / mol L^{-1} min^{-1}**
I	0.1	0.1	2.0×10^{-2}
II	–	0.2	4.0×10^{-2}
III	0.4	0.4	–
IV	–	0.2	2.0×10^{-2}

E13 Find the half-life of a first order reaction whose rate constant is (i) 200 s^{-1}; (ii) 2 min^{-1}; (iii) 4 years^{-1}.

E14 ^{14}C decays radioactively with a half-life of 5730 years. A wooden archaeological artifact holds only 80% of the ^{14}C present in a living tree. Estimate its age.

E15 Gas-phase decomposition of N_2O_5 [$2N_2O_5 \rightarrow 4NO_2 + O_2$] at 318 K gave:

t / s	0	400	800	1200	1600	2000	2400	2800	3200
$10^2 \times$ [N_2O_5] / mol L^{-1}	1.63	1.36	1.14	0.93	0.78	0.64	0.53	0.43	0.35

(i) Plot [N_2O_5] against t. (ii) Find the half-life. (iii) Plot log[N_2O_5] against t. (iv) What is the rate law?

Exercises continued

(E15 continued) (v) Work out the rate constant. (vi) Find the half-life from k and compare it with your answer to (ii).

E16 A first order reaction has $k = 60\ s^{-1}$. How long does it take for the reactant's concentration to fall to 1/16th of its initial value?

E17 ^{90}Sr, a product of nuclear explosions, has a half-life of 28.1 years. If a newborn baby's bones take up 1 mg of ^{90}Sr in place of calcium, and none is lost metabolically, how much will be left after 10 years and after 60 years?

E18 Show that a first order reaction takes twice as long to reach 99% completion as to reach 90% completion.

E19 A first order reaction is 30% decomposed in 40 min. Find $t_{1/2}$.

E20 Azoisopropane decomposes to hexane and nitrogen at 543 K. From the data, find the rate constant.

t (sec)	0	360	720
P (mm of Hg)	35.0	54.0	63.0

E21 SO_2Cl_2 decomposes thermally by first order kinetics at constant volume: $SO_2Cl_2(g) \rightarrow SO_2(g) + Cl_2(g)$. Total pressure is 0.5 atm at 0 s and 0.6 atm at 100 s. What is the rate when the total pressure is 0.65 atm?

E22 Rate constants for the decomposition of N_2O_5 at several temperatures are given. Plot ln k against $1/T$, find A and E_a, and predict k at 30 °C and 50 °C.

T / °C	0	20	40	60	80
$10^5 \times k$ / s^{-1}	0.0787	1.70	25.7	178	2140

E23 At 546 K the rate constant for decomposition of hydrocarbons is $2.418 \times 10^{-5}\ s^{-1}$, and the activation energy is 179.9 kJ/mol. Find the pre-exponential factor.

E24 For A → Products, $k = 2.0 \times 10^{-2}\ s^{-1}$. Starting from 1.0 mol L^{-1} of A, how much A is left after 100 s?

E25 Sucrose in acid solution splits into glucose and fructose, following first order kinetics with $t_{1/2} = 3.00$ hours. What fraction of a sucrose sample is left after 8 hours?

E26 A hydrocarbon decomposes according to $k = (4.5 \times 10^{11}\ s^{-1})\ e^{-28000\ K/T}$. Find E_a.

Exercises continued

E27 For the first order decomposition of H_2O_2, $\log k = 14.34 - 1.25 \times 10^4$ K/T. Find E_a, and the temperature at which the half-period is 256 minutes.

E28 For the decomposition of A into product, $k = 4.5 \times 10^3$ s^{-1} at 10 °C and $E_a = 60$ kJ mol^{-1}. At what temperature is $k = 1.5 \times 10^4$ s^{-1}?

E29 A first order reaction takes as long to reach 10% completion at 298 K as to reach 25% completion at 308 K. Given $A = 4 \times 10^{10}$ s^{-1}, find k at 318 K and E_a.

E30 A reaction's rate becomes four times greater when the temperature goes from 293 K to 313 K. Assuming E_a does not vary with temperature, find it.

Answers to some intext questions

Question	Answer
Q1	$r_{av} = 6.66 \times 10^{-6}$ M s^{-1}
Q2	rate of reaction = rate of disappearance of A = 0.005 mol litre^{-1} min^{-1}
Q3	the order of the reaction is 2.5
Q4	X → Y, Rate = k[X]^2; the rate becomes 9 times greater
Q5	$t = 444$ s
Q6	1.925×10^{-4} s^{-1}
Q8	$E_a = 52.897$ kJ mol^{-1}
Q9	1.471×10^{-19}